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Let us index the trees from 1 to 6. The index of a sparrow is the index of the tree it is currently sitting on. A pair of sparrows flying can be modeled as an iterative step of a loop. When a sparrow at tree i flies to tree i + d, another sparrow at tree j flies to tree j — d. Thus, after each iterative step, the sum S of the indices of the sparrows remains invariant. Moreover, a loop invariant is true at the start and at the end of the loop.

At the start of the loop, the value of the invariant is

S = 1 + 2 + 3 + 4 + 5 + 6 = 21

When the loop ends, the loop invariant has the same value. However, when the loop ends, if all the sparrows were on the same tree, say k, then S = 6k.

S = 21,	loop invariant at the start of the loop
S = 6k,	loop invariant at end of the loop
6k= 21,	loop invariant has the same value at the start and the end
21 is a multiple of 6	

It is not possible — 21 is not a multiple of 6. The desired final values of the sparrow indices is not possible with the loop invariant. Therefore, all the sparrows cannot gather on one tree.

Example 8.8. Consider the Chameleons of Chromeland of Example 6.3. There are 13 red, 15 green, and 17 blue chameleons on Chromeland. When two chameleons of different colors meet they both change their color to the third one (for example, if a red and a green meet, both become blue). Is it possible to arrange meetings that result in all chameleons displaying blue color?

Let r, g, and b be the numbers of red, green and blue chameleons. We can model the meetings of two types as an iterative process. A meeting changes (r, g, b) into (r-1, g-1, b+2) or (r-1, g+2, b-1) or (r+2, g-1, b-1). Consider, for example, the meeting of a red and a green chameleon.

r, g, b := r-1, g-1, b+2

The difference in the numbers of any two types either do not change or changes by 3. This is an invariant.

r - 1 - (g - 1) = r - g

r - 1 - (b + 2) = (r - b) - 3

g - 1 - (b + 2) = (g - b) - 3

This is true for all three cases. If any two types differ in number by a multiple of 3 at the start of the iterative process, the difference can be reduced in steps of 3, to 0, when the iterative process ends. However, at the start,

r - g = 13 - 15 = -2

g - b = 15 - 17 = -2

g - r = 17 - 13 =4

No two colors differ in number by a multiple of 3. Therefore, all the chameleons cannot be changed to a single color.